

PointMetricsTest

Processed Using

Lapis Version 0.9

July 22, 2025

RunParameters Folder

The RunParameters folder contains files that have the full details of this run, which can be loaded into Lapis to duplicate part or all of this run.

FullParameters.ini

FullParameters.ini contains all parameters used in this run. Loading it will allow you to duplicate it precisely.

DataParameters.ini

DataParameters.ini contains the parameters that describe the input data of the run. Loading it will allow you to process the same data with new options.

ProcessingParameters.ini

ProcessingParameters.ini contains the parameters that describe the method by which the data was processed. Loading it will allow you to process a new dataset with the same methods.

ComputerParameters.ini

ComputerParameters.ini contains the parameters which are specific to the computer on which the run was performed. Loading it will allow you to perform runs on the same computer without needing to re-set them.

ProcessingAndComputerParameters.ini

ProcessingAndComputerParameters.ini contains all parameters in either ComputerParameters.ini or ProcessingParameters.ini. It's like a profile for how you process lidar, which you can apply to new datasets.

Point Filters

It is very common for lidar data files to include a large number of points which are unsuitable for processing, either because they are errors or because you're doing a special run. This page describes the strategies used to exclude those points from the run.

Outlier Filter

Points below -8 meters or above 100 meters were excluded.

Withheld Filter

Points are sometimes marked 'withheld', indicating that a previous piece of software concluded they were unsuitable for use. Withheld points were excluded from this run.

Class Filter

Lapis does not perform point classification. However, it is common for input data to already be classified by some previous software. If this is the case for this data, then the following standard classes were filtered:

Low Point

Water

High Noise

Scan Angle Filter

Aerial lidar sensors sweep back and forth as they emit pulses. The angle that a pulse was fired at is called the scan angle for that pulse. For some applications, points with a high scan angle are undesirable. In this run, points with a scan angle greater than 30 degrees were excluded.

Output Data Characteristics

All metrics were processed at a cellsize of 10 meters.

Where appropriate, the units of all output data are meters.

All output data of this run is in the following coordinate reference system: WGS 84 / UTM zone 10N + NAVD88 height. The full WKT string is provided below for reference:

```
COMPOUNDCRS["WGS 84 / UTM zone 10N + NAVD88 height",
  PROJCRS["WGS 84 / UTM zone 10N",
    BASEGEOGCRS["WGS 84",
      ENSEMBLE["World Geodetic System 1984 ensemble",
        MEMBER["World Geodetic System 1984 (Transit)"],
        MEMBER["World Geodetic System 1984 (G730)"],
        MEMBER["World Geodetic System 1984 (G873)"],
        MEMBER["World Geodetic System 1984 (G1150)"],
        MEMBER["World Geodetic System 1984 (G1674)"],
        MEMBER["World Geodetic System 1984 (G1762)"],
        MEMBER["World Geodetic System 1984 (G2139)"],
        MEMBER["World Geodetic System 1984 (G2296)"],
        ELLIPSOID["WGS 84",6378137,298.257223563],
        ENSEMBLEACCURACY[2.0]],
      UNIT["degree",0.0174532925199433],
      ID["EPSG",4326]],
    CONVERSION["UTM zone 10N",
      METHOD["Transverse Mercator"],
      PARAMETER["Latitude of natural origin",0],
      PARAMETER["Longitude of natural origin",-123],
      PARAMETER["Scale factor at natural origin",0.9996],
      PARAMETER["False easting",500000],
      PARAMETER["False northing",0]],
```

CS[Cartesian,2],

AXIS["(E)",east],

AXIS["(N)",north],

UNIT["metre",1],

USAGE[

SCOPE["Navigation and medium accuracy spatial referencing."],

AREA["Between 126°W and 120°W, northern hemisphere between equator and 84°N, onshore and offsl

BBOX[0,-126,84,-120]],

ID["EPSG",32610]],

VERTCRS["NAVD88 height",

VDATUM["North American Vertical Datum 1988"],

CS[vertical,1],

AXIS["gravity-related height (H)",up],

UNIT["metre",1],

USAGE[

SCOPE["Geodesy, engineering survey, topographic mapping."],

AREA["Mexico - onshore. United States (USA) - CONUS and Alaska - onshore - Alabama; Alaska; Ariz

BBOX[14.51,172.42,71.4,-66.91]],

ID["EPSG",5703]]]

Ground Model Algorithm

In this run, Lapis did not calculate its own ground model. Instead, the user supplied a rasterized ground model produced by previous software. The elevation of the ground beneath each point was estimated with a bilinear interpolation from those rasters.

Point Metrics

The PointMetrics folder contains summary statistics calculated on the height values of lidar returns contained in each cell of the raster. They are commonly used as inputs to models, and some are interpretable values on their own. In many of them, returns below the canopy are excluded. It is important for most purposes to exclude ground returns from these metrics, but by adjusting the definition of canopy, you can exclude or include shrubs. In this run, canopy points are defined as points at least 1.5 meters above the ground. Two versions of each metric exist: one calculated using only first returns, and one calculated with all returns. The first return versions are in the PointMetrics/FirstReturns directory, and the all return versions are in the PointMetrics/AllReturns directory.

PointMetricsTest_XXthPercentile_CanopyHeight_Meters.tif

There are a large number of metrics of this form, replacing XX with a specific number. They are calculated as the given percentile of the heights of canopy returns. The 25th percentile is a proxy for height to live crown. The 95th percentile is a proxy for the height of the tallest tree. They are measured in meters.

PointMetricsTest_Mean_CanopyHeight_Meters.tif

The mean height of all canopy returns. The units are meters.

PointMetricsTest_StdDev_CanopyHeight_Meters.tif

The standard deviation of height of all canopy returns. The units are meters.

PointMetricsTest_TotalReturnCount.tif

The total number of returns. Mostly useful for data validation. This metric is unitless.

PointMetricsTest_CanopyCover_Percent.tif

The percentage of returns which are canopy returns. A proxy for canopy cover. The units are percent.

PointMetricsTest_CoverAboveMean_Percent.tif

The percentage of returns with a height greater than the mean height. The units are percent.

PointMetricsTest_CanopyReliefRatio.tif

A combination of the mean, minimum, and maximum values of return height. The formula is $(\text{mean} - \text{min}) / (\text{max} - \text{min})$. This metric is unitless.

PointMetricsTest_CanopySkewness.tif

The mathematical concept of skewness, calculated on the height of canopy returns. A measure of how symmetric the canopy profile is around the center of the canopy. A high value indicates a lack of symmetry. This metric is unitless.

PointMetricsTest_CanopyKurtosis.tif

The mathematical concept of kurtosis, calculated on the height of canopy returns. A measure of how spread the canopy profile is. A high value indicates a lot of spread. This metric is unitless.

PointMetricsTest_Mean_Intensity.tif

The mean intensity of all points in the cell. Useful for assessing at-a-glance whether intensity values are comparable across the entire area. This metric is unitless.

Stratum metrics

Some metrics are calculated on vertical slices of the lidar returns. Their filename will indicate what slice was used to calculate them. These metrics exist in the StratumMetrics directory, under the point metrics directory. The strata used in this run are:

LessThan0.5Meters

0.5To5.5Meters

5.5To10.5Meters

GreaterThan10.5Meters

PointMetricsTest_StratumCover_XXtoXX_Percent.tif

The number of returns that fall in this stratum, as a percentage of the number of returns in this stratum or lower. A proxy for the cover present in this stratum. The units are percent.

PointMetricsTest_StratumPercent_XXtoXX_Percent.tif

The number of returns that fall in this stratum, as a percentage of the total number of returns. The units are percent.